TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Development of a High-Performance Safety Helmet Based on Radiation-Shielding Metal-Yarn Fabric Insert Over-Molding for Nuclear Power Plant Workers			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power (√), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and Materials		***Energy & Resources***	
	Medium Category	Small Category	Medium Category	Small Category
	Polymeric Materials	*Composite Material Manufacturing Technologies*	*Nuclear Power*	*Radiation Management Technologies*
Concept of Proposed Technology	**Concept of Proposed Technology** **(Concept)** The proposed technology aims to develop a core manufacturing technology in which **tungsten metal-yarn fabric is inserted into plastic components through an insert over-molding process**, enabling the **integrated molding of metal yarn and polymer**. The technology targets a range of safety and protective components used in nuclear power plant environments, including **safety helmets as the primary application**, as well as shielding panels, gloves, and cable covers. **(Technical Challenges)** Conventional metal insert molding technologies mainly rely on **adhesives or mechanical fastening methods**, which often lead to **reduced bonding strength under high-temperature and high-humidity conditions**, such as those encountered in nuclear power plants. In			

	addition, these approaches have limitations in achieving stable and durable integration between metal inserts and plastic components, making them less suitable for long-term use in demanding operational environments. **(International Joint R&D Approach)** In this project, **advanced plasma-based surface treatment technologies for metal-yarn fabrics**, developed by leading overseas research institutes such as Fraunhofer, are applied to **enhance the interfacial bonding performance between metal yarns and polymers**. Based on this technology, domestic partners focus on **optimizing insert over-molding process conditions** applicable to components with various geometries, thereby ensuring practical applicability in actual product manufacturing. **(Key Functions and Technical Features)** Tungsten, with an **atomic number of 74 and a density of 19.25 g/cm³**, is a high–atomic-number, high-density material that provides **gamma-ray and X-ray shielding performance comparable to lead**, while being environmentally friendly. By configuring tungsten in the form of a **woven metal-yarn fabric** and integrating it with polymers such as ABS through insert over-molding, the proposed technology enables **radiation shielding and flame-retardant functionality**, while maintaining **mechanical strength and wearability comparable to conventional plastic components**. **(Applications and Expandability)** The proposed technology can be primarily applied to **safety equipment for nuclear power plant workers**, including safety helmets, shielding panels, tool containers, and cable covers. Furthermore, it can be extended to a wide range of industries requiring **high-performance and lightweight composite structures**, such as **radiation inspection and medical shielding equipment, aerospace and space structures, electronic component encapsulation, and electromagnetic shielding products**.
Rationale for Joint R&D Project	This international joint R&D project aims to **localize the manufacturing of radiation-shielding safety helmets and protective components applicable to nuclear power plant environments**. The project is based on the necessity to **overcome technical limitations in metal-yarn surface treatment and insert over-molding core processes**, which are difficult to secure through domestic efforts alone. Currently, there is **limited domestic experience and know-how** in surface treatment technologies that enable the **stable integration of tungsten metal-yarn fabrics with plastics**, as well as in **insert over-molding processes specifically tailored for metal-yarn fabrics**. Conventional approaches relying on adhesives, or mechanical fastening often suffer from **reduced bonding performance under high-temperature and high-humidity conditions**, such as those encountered in nuclear power plant environments, thereby limiting their suitability for long-term use in radiation-shielding components. Tungsten-powder filled compounds for injection molding are already commercially available. However, for a helmet or further protective equipment application, there are high mechanical requirements leading to the need for endless fiber reinforcement. Internationally, **radiation shielding technologies utilizing high-atomic-number metals** have been actively studied, with particular emphasis on **plasma-based surface treatment and interface control technologies for metal-yarn fabrics**. Leading overseas research institutes, such as Fraunhofer in

	Germany, possess **advanced core technologies in metal-yarn surface treatment and insert over-molding process design**, which are difficult to establish domestically within a short time frame. Accordingly, this project seeks to **jointly develop and adapt these core technologies through international collaboration**, and to link them with the **mold design, injection molding, and product manufacturing capabilities of Korean SMEs**, thereby **internalizing manufacturing technologies for radiation-shielding safety helmets and protective components suitable for nuclear-specific operating environments**. Domestic research institutes support the project by **verifying process conditions and evaluating performance**, ensuring technical reliability. Through this international joint R&D effort, the project is expected to achieve **localization of core manufacturing processes**, **improvement of technology readiness through prototype fabrication and performance validation**, and further **expansion of applications to nuclear power plant maintenance, decommissioning, and inspection activities**.
Key Words	1. Tungsten Metal Yarn Fabric 2. Insert Over-Molding 3. Radiation Shielding PPE 4. Metal–Polymer Interface Engineering 5. Nuclear Power Plant Safety Equipment

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (7)
R&D Objectives	The objective of this R&D project is to **secure a core manufacturing technology that combines tungsten metal-yarn fabrics with an insert over-molding process** for radiation-shielding safety helmets and protective components applicable to the **special operating environments of nuclear power plants**, and to **advance the technology readiness to a prototype level suitable for practical application**. To this end, the project will be carried out in a stepwise manner, covering **material development, surface treatment, process optimization, prototype fabrication, and performance validation**. • Development of Metal-Yarn Surface Treatment Technologies : Develop woven structures using lightweight, high-density tungsten metal yarns, and secure both radiation-shielding performance and flexibility through plain or twill fabric designs and fabric surface coating technologies. • Optimization of the Insert Over-Molding Process : Establish mold designs and injection molding conditions applicable to various component geometries—including safety helmets, shielding panels, gloves, and cable covers—to improve the reproducibility and productivity of the insert over-molding process. • Prototype Fabrication and Performance Evaluation : Fabricate prototypes of safety helmets and shielding components incorporating tungsten metal-yarn fabrics, and conduct radiation shielding efficiency, mechanical strength, thermal resistance, and required property tests to validate performance and build a structured performance database.			

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	• Technology Securing and Industrialization Foundation : Secure the developed metal-yarn surface treatment and insert over-molding technologies through joint patents and publications in collaboration with Fraunhofer ICT, and establish design and manufacturing databases to support the commercialization of radiation-shielding products for nuclear power plant applications.
R&D Target Specifications	• 1. Thickness of the Tungsten fiber : 0.2 millimeter ↓ • 2. Thickness of the fabric: 0.3 millimeter ↓ • 3. Thickness of the shielding material : 1 millimeter ↓ • 4. Lead equivalent : 0.1 mmPb ↑ • 5. Flame Resistance-Burning time : 5sec ↓ • 6. Shock Absorption-Peak Transmitted Force : 4,450N ↓
R&D Plan	**Phase 1 (Basic Research Phase, 6 months)** • **Dongwon Entech (Korean SME Lead Organization)** - Conduct market analysis and review relevant standards for application in nuclear power plant environments, and select target products in collaboration with end users (nuclear power plant operators). - Establish basic mold design concepts for the selected products. • **Fraunhofer ICT (International Partner)** - Review trends in surface treatment technologies for the application of tungsten metal-yarn fabrics, and analyze the applicability of plasma-based surface treatment technologies suitable for insert over-molding processes. - Development of concepts for fixing a woven metal-yarn fabric insert in a mold during the over-molding process • **KTDI (Contracting Research Institute)** - Select key specifications of tungsten metal yarns, including diameter and grade. - Establish design concepts for high-density woven fabric structures, such as plain and twill weaves. • **UNIST (Contracting Research Institute)** - Conduct basic research on injection molding processes considering metal-yarn fabric inserts, and evaluate the feasibility of applying insert over-molding processes. **Phase 2 / Technology Development, Validation, and Prototype Fabrication Phase (30 months)** • **Dongwon Entech (Korean SME Lead Organization)** - Design and fabricate molds for nuclear power plant safety-related products, including safety helmets, gloves, and cable covers. - Fabricate prototypes using the insert over-molding process. - Conduct final product performance evaluation, analysis, and support relevant certification procedures. • **Fraunhofer ICT (International Partner)** - Perform plasma-based surface treatment experiments on tungsten metal-yarn fabrics and derive conditions for improved interfacial bonding performance. - Develop a robot-assisted gripper system for applying insert over-molding processes. - Optimize the insert over-molding process through repeated experiments and transfer core process technologies. - Identify critical design criteria and communication to mold maker in Korea. - Advisory role in design of products and further process optimization. - Environmental impact assessment with LCA screening for treatment process and radiation shielding safety helmet. • **KTDI (Contracting Research Institute)** - Design and study manufacturing processes for high-density tungsten metal-yarn fabric weaving. - Conduct process optimization studies for fabric structures, develop metal-yarn fabrics, and stabilize the weaving process. • **UNIST (Contracting Research Institute)** - Conduct both basic and advanced studies on metal-yarn fabric insert over-molding processes.

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- Support the internalization of technologies developed by overseas research institutes and provide technical support for molding process implementation.

- **Phased R&D Implementation Plan and SMART Milestones**

Detailed Task		Institution	Country	Phase 1		Phase 2 1st Year				Phase 2 2nd Year				Phase 2 3rd Year	
				Q1	Q2	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
WP1. Conceptual Design of Radiation-Shielding Safety Products for Workers															
1-1	Investigation of nuclear environment & standards, selection of target products	Dongwon Entech	KOR												
1-2	Selection of tungsten metal-yarn specifications	KTDI	KOR												
1-3	Design of high-density woven fabric structures	KTDI	KOR												
1-4	Conceptual structural design of target products	Dongwon Entech	KOR												
1-5	Conceptual study on feasibility of fabric insert over-molding	Fraunhofer ICT UNIST	GER KOR												
1-6	Review of surface treatment technology directions	Fraunhofer ICT	GER												
WP2. Development of Metal-Yarn, Surface Treatment, and Over-Molding Processes															
2-1	Design of fabric weaving and coating processes	KTDI	KOR												
2-2	Evaluation of fabric properties and radiation-shielding performance	KTDI	KOR												
2-3	Surface treatment experiments for metal-yarn fabrics	Fraunhofer ICT	GER												
2-4	Derivation of insert over-molding process conditions	Fraunhofer ICT	GER												
2-5	Property testing in research level	Fraunhofer ICT	GER												
2-6	Basic injection molding experiments	UNIST	KOR												
2-7	Insert automation (gripper) concept development with commercial fabrics	Fraunhofer ICT	GER												
2-8	Insert automation (gripper) concept development with developed fabric	Fraunhofer ICT	GER												
WP3. Prototype Fabrication and Repetitive Molding Experiments															
3-1	Fabrication and support of insert over-molding molds	Dongwon Entech UNIST	KOR												
3-2	Product manufacturing processes & post-processing support	Dongwon Entech UNIST	KOR												
3-3	Fabrication of worker safety product prototypes	Dongwon Entech	KOR												
3-4	Evaluation of process reproducibility	Dongwon Entech	KOR												
3-5	Technical advisory on process parameter and mold design risk identification	Fraunhofer ICT	GER												
3-6	Improvement of high-density tungsten fabric structures based on prototype feedback	KTDI	KOR												
3-7	Optimization of fabric design considering shielding performance and moldability	KTDI	KOR												
3-8	LCA screening for the surface treatment and radiation shielding safety helmet	Fraunhofer ICT	GER												
WP4. Process Optimization and Final Validation															
4-1	Radiation-shielding performance testing	Dongwon Entech	KOR												

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4-2	Mechanical performance testing	Dongwon Entech	KOR													
4-3	Optimization of high-density metal-yarn weaving processes	KTDI	KOR													
4-4	Process optimization & technical support for prototypes	Dongwon Entech UNIST Fraunhofer ICT	KOR GER													
4-5	Final performance validation and certification	Dongwon Entech	KOR													
4-6	Technology documentation and reporting	Dongwon Entech UNIST KTDI Fraunhofer ICT	KOR GER													

- **Detailed R&D Work Plan and Schedule by Work Package**

Phase 1 (6 month)

Phase 1) Concept Design of Radiation-Shielding Safety Products

Dongwon Entech	
WP 1-1	Nuclear environment & standards analysis
WP 1-4	Safety product structural concept design

KTDI	
WP 1-2	Tungsten metal-yarn specification definition
WP 1-3	High-density woven fabric structure design

Fraunhofer ICT	
WP 1-5	Conceptual study on feasibility of fabric insert over-molding
WP 1-6	Surface treatment technology direction review

UNIST	
WP 1-5	Conceptual study on feasibility of fabric insert over-molding

Phase 2 1st Year (12 month)

Phase 2-1) WP2. Metal-Yarn & Over-Molding Process Development (Early Stage) + WP3. Early-Stage Prototype Verification

KTDI	
WP 2-1	Fabric weaving & coating process design
WP 2-2	Fabric properties & radiation-shielding performance evaluation

Fraunhofer ICT	
WP 2-3	Metal-yarn surface treatment experiments
WP 2-4	Derivation of insert over-molding process conditions
WP 2-7	Insert automation (gripper) concept development with commercial fabric
WP 2-8	Insert automation (gripper) concept development with developed fabric

UNIST	
WP 2-6	Basic injection molding experiments

Dongwon Entech	
WP 3-1	Preliminary fabrication and support of insert over-molding molds
WP 3-3	Fabrication of early-stage worker safety product prototypes
WP 3-4	Initial evaluation of process reproducibility under limited conditions

Phase 2 2nd Year (12 month)

Phase 2-2) WP3. Prototype Fabrication & Process Refinement

Dongwon Entech	
WP 3-2	Product manufacturing process refinement & post-processing support
WP 3-3	Fabrication of worker safety product prototypes

UNIST	
WP 3-2	Injection molding & post-processing technical support
WP 3-4	Process stability and manufacturability verification

Fraunhofer ICT	
WP 3-5	Technical advisory on process parameters and mold design risk identification
WP 3-8	LCA for the treatment process and prototype

KTDI	
WP 3-6	Optimization of high-density tungsten fabric structures based on prototype feedback
WP 3-7	Improvement of tungsten fabric design considering radiation-shielding performance and moldability

Phase 2 3rd Year (6 month)

Phase 2-3) WP4. Final Validation, Optimization & Technology Transfer

Dongwon Entech	
WP 4-1	Application environment analysis
WP 4-2	Hybrid chock liner structural design
WP 4-6	Technology documentation & reporting

KTDI	
WP 4-3	Optimization of high-density metal-yarn weaving process
WP 4-6	Technology documentation & reporting

Fraunhofer ICT	
WP 4-4	Prototype process optimization & technical support
WP 4-6	Technology documentation & reporting

UNIST	
WP 4-4	Prototype process optimization & technical support
WP 4-6	Technology documentation & reporting

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**International Research Institute: Fraunhofer ICT** • Fraunhofer ICT serves as the key overseas research institute leading the development of core insert over-molding process technologies based on tungsten metal-yarn fabrics in this international joint R&D project. • Leveraging its expertise in composite forming and surface engineering, Fraunhofer ICT jointly develops core technologies for applying insert over-molding processes, focusing on the analysis of metal-yarn–polymer interfacial characteristics and plasma-based surface treatment technologies. - Analysis of interfacial bonding mechanisms between tungsten metal-yarn fabrics and plastics, and establishment of related technical concepts - Derivation of surface treatment conditions for metal-yarn fabrics applicable to insert over-molding - Review of process applicability and provision of design guidelines for insert over-molding processes - Documentation of core process technologies and transfer of know-how to domestic institutions • Through these activities, Fraunhofer ICT acts as a technical core partner supporting the acquisition of key manufacturing process technologies and their internalization in Korea through international joint research. **Domestic Research Institute: KTDI** • KTDI serves as a specialized domestic research institute responsible for the development of tungsten metal yarns and woven fabrics, as well as the evaluation of radiation-shielding performance in this project. • Based on its expertise in metal-yarn fabric design and performance evaluation, KTDI verifies the technical feasibility of radiation-shielding structures. - Selection of tungsten metal-yarn specifications and design of high-density woven fabric structures - Research on weaving processes for metal-yarn fabrics and development of fabric materials - Evaluation of radiation-shielding performance and mechanical properties of fabrics and prototypes - Organization of test data and technical support for certification-related activities **Domestic Research Institute: UNIST** • UNIST serves as a domestic research institute responsible for the engineering verification and internalization of metal-yarn fabric insert over-molding injection molding processes in this project. • By conducting technical verification from a molding-process perspective, UNIST supports product development by the lead organization. - Basic research on injection molding processes incorporating metal-yarn fabric inserts - Analysis of molding behavior and quality characteristics under varying injection conditions - Verification of process stability and reproducibility through repeated molding trials - Application of molding processes and technical support during prototype fabrication - Optimization of molding processes in the later stage of the project to establish process conditions suitable for repeated application •
Expected R&R of Korean SME Partner	**Korean SME Lead Organization: Dongwon Entech** • Dongwon Entech serves as the Korean SME lead organization responsible for overseeing the development of radiation-shielding safety helmets and protective components for nuclear power plant applications in this international joint R&D project. • The company integrates and manages the entire R&D process, with a focus on product planning, mold and manufacturing process design, prototype fabrication, and performance validation. - Definition of product requirements and selection of target products considering nuclear power plant operating environments - Product structural design and mold design/fabrication considering insert over-molding applications

	- Fabrication of prototypes, including safety helmets, gloves, and cable covers, and verification of repeatable production feasibility - Product design refinement and final validation based on radiation-shielding and mechanical performance evaluation results - Establishment of commercialization strategies and support for certification processes based on project outcomes • Through these activities, Dongwon Entech acts as the core implementation entity that translates technological outcomes from overseas and domestic research institutes into actual products and manufacturing processes, leading the commercialization of radiation-shielding products for nuclear power plant applications as a Korean SME.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Technical Outcomes** • Securing radiation-shielding materials and component technologies based on environmentally friendly tungsten metal-yarn fabrics as an alternative to lead (Pb)-based shielding materials. • Establishment of metal–polymer hybrid manufacturing technologies based on insert over-molding, enabling the integrated molding of metal-yarn fabrics and plastics. • Development of prototypes of radiation-shielding products—including safety helmets, shielding panels, gloves, and cable covers—for enhancing worker safety in nuclear power plant environments, along with the establishment of a basis for relevant standards and certification. • Creation of joint patents and publications through international collaboration on core manufacturing process technologies, thereby strengthening the technological self-reliance and design/manufacturing capabilities of domestic companies. **Expected Industrial and Economic Impact** • Localization of shielding components and products used in nuclear power plants and radiation-related facilities, leading to import substitution and the establishment of a foundation for public-sector procurement. • Creation of new markets through the expansion of radiation-shielding safety equipment and protective product lines utilizing tungsten metal-yarn fabrics and over-molding technologies. • Securing opportunities for expanded applications in high value-added industries such as medical, energy, aerospace, and space sectors, with nuclear applications as the initial focus. • Enhancement of technological credibility and industrial partnerships of SMEs based on domestic and international performance validation and certification. **Expected Impact on Global Collaboration** • Achievement of joint development and domestic internalization of core manufacturing process technologies, rather than simple technology transfer, through collaboration with overseas research institutes. • Strengthening of the capabilities of domestic companies and research institutes in metal–polymer hybrid manufacturing technologies and radiation-shielding applications. • Establishment of a foundation for expanding international joint R&D into safety equipment for special operating environments, including the nuclear industry. **Expected Societal and Strategic Impact** • Contribution to enhanced worker safety and reduced radiation exposure in nuclear power plant maintenance, decommissioning, and inspection operations. • Promotion of a technological transition from conventional plastic-based safety equipment to high-performance, environmentally friendly composite-based safety solutions. • Strengthening of the competitiveness of the nuclear equipment industrial ecosystem through SME-led technology development and international joint research.

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	• Provision of a technological foundation for expanding radiation-shielding technologies into strategic industries such as medical diagnostics, inspection, aerospace, and space applications.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization